



**Politecnico
di Torino**

Kinematic Behavior of a Vehicle Front Suspension

Bachelor's Degree in Automotive Engineering

Loris Fonseca: `s295208@studenti.polito.it`

Antonio Giuliani: `s295452@studenti.polito.it`

Franco Porras Yarasca: `s303056@studenti.polito.it`

Karim Abdelgelil Mohamed Shalaby: `s302311@studenti.polito.it`

a.y 2023/2024

Abstract

The aim of this project is to study the static and kinematic behavior of a double wishbone suspension of a vehicle with traction on the front axle. For this purpose, a 3D model of the system is built on Adams Car and the same software is then used to simulate the movements during vertical wheel travel. Then another simulation is run considering a different set of hardpoints coordinates, to see how the results are affected. In the final part, the previous results are used to design an appropriate bump-stop for the shock absorber.

Contents

1	Introduction	4
2	Model of the suspension	4
3	Kinematic behavior	4
4	Sensitivity of the kinematic behavior to a change in the hard-points position	5
5	Static forces on the suspension	6
6	Conclusions	9
A	Appendix	10

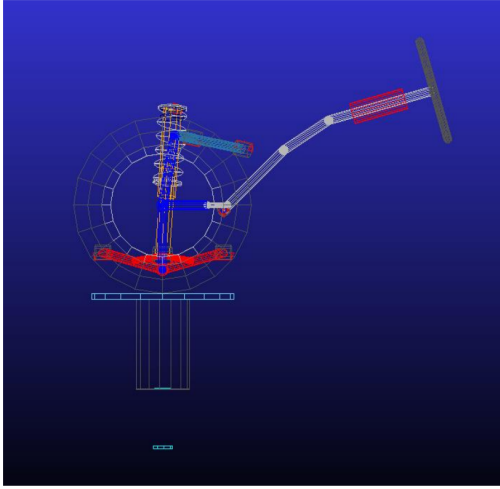
1 Introduction

The first task is to build a model of the suspensions on Adams Car, based on the given hardpoints. The front suspension of the Alfa Romeo 156 is used as reference. Its kinematic behavior is simulated, giving as input the wheel travel. The whole process is then repeated with a slightly different set of hardpoints.

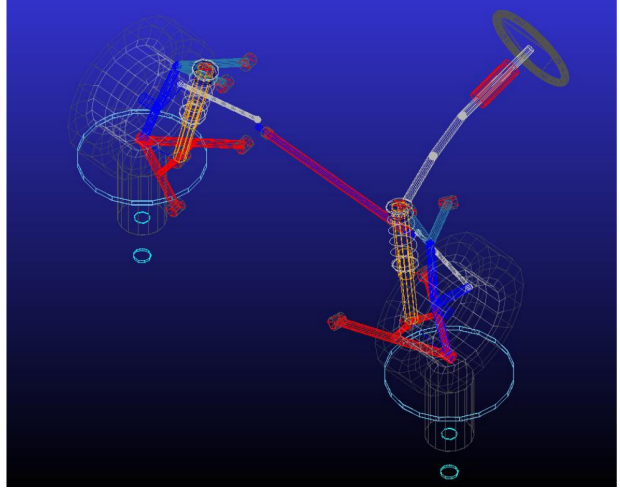
The second task is to evaluate the static forces and stiffness of the suspension which imply the evaluation of the vertical load on the ground for different suspension travel positions in bound condition and in rebound condition. In addition it is requested to design the bump-stops, and to determine the wheel-ground stiffness.

2 Model of the suspension

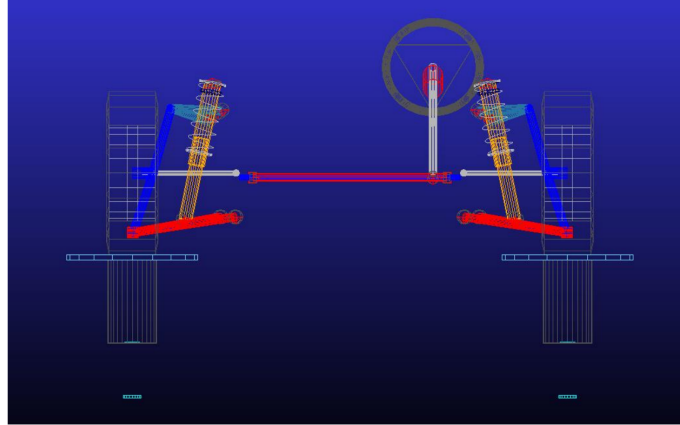
To build the model, the default geometries of Adams Car are modified to match the hardpoints, then camber and toe angles are set to zero and the correct tire unloaded radius is selected.



(a) Suspension side view - normal conditions



(b) Suspension top view - normal conditions



(c) Suspension front view - normal conditions

Figure 1: Suspension views in normal conditions

3 Kinematic behavior

The kinematic behavior is simulated according to the wheel travel, imposed as maximum bump-travel of 60 mm and a maximum rebound-travel of -90 mm. Relevant parameters are tabulated and plotted for better visualization in the green table in Figure 2.

4 Sensitivity of the kinematic behavior to a change in the hard-points position

The position of the upper and lower wishbones frame mounts are modified to evaluate the sensitivity of the kinematic behavior. In detail, the lower control arm is raised by 10 mm and the upper control arm is lowered by the same amount. Another simulation is run and the results are compared with the previous ones.

Reference hardpoints					Modified hardpoints				
Wheel Travel	Toe angle	Camber angle	Semi-Track length	Track length	Wheel Travel	Toe angle	Camber angle	Semi-Track length	Track length
z	α_0	γ_0	$t_0/2$	t_0	z	α_0	γ_0	$t_0/2$	t_0
(mm)	(deg)	(deg)	(mm)	(mm)	(mm)	(deg)	(deg)	(mm)	(mm)
-90.00	7.84	-0.80	758.28	1516.56	-90.00	7.86	-0.82	758.37	1516.75
-84.00	6.96	-0.65	761.14	1522.27	-84.00	6.99	-0.67	761.21	1522.43
-78.00	6.16	-0.52	763.82	1527.64	-78.00	6.18	-0.53	763.89	1527.77
-72.00	5.43	-0.41	766.35	1532.70	-72.00	5.44	-0.42	766.40	1532.80
-66.00	4.74	-0.32	768.72	1537.44	-66.00	4.76	-0.32	768.76	1537.52
-60.00	4.12	-0.24	770.95	1541.90	-60.00	4.12	-0.24	770.98	1541.96
-54.00	3.53	-0.17	773.04	1546.07	-54.00	3.54	-0.17	773.06	1546.11
-48.00	3.00	-0.11	774.99	1549.98	-48.00	3.00	-0.11	775.00	1550.00
-42.00	2.50	-0.06	776.81	1553.62	-42.00	2.50	-0.06	776.82	1553.63
-36.00	2.04	-0.03	778.50	1557.00	-36.00	2.04	-0.03	778.50	1557.01
-30.00	1.61	0.00	780.07	1560.14	-30.00	1.61	0.00	780.07	1560.14
-24.00	1.22	0.02	781.52	1563.04	-24.00	1.22	0.02	781.52	1563.03
-18.00	0.86	0.02	782.85	1565.70	-18.00	0.86	0.02	782.85	1565.70
-12.00	0.53	0.02	784.07	1568.14	-12.00	0.53	0.02	784.07	1568.14
-6.00	0.23	0.00	785.18	1570.35	-6.00	0.23	0.00	785.18	1570.35
0.00	-0.04	-0.02	786.17	1572.35	0.00	-0.04	-0.02	786.18	1572.35
6.00	-0.29	-0.05	787.06	1574.13	6.00	-0.29	-0.05	787.07	1574.14
12.00	-0.51	-0.09	787.85	1575.70	12.00	-0.51	-0.09	787.86	1575.72
18.00	-0.70	-0.14	788.53	1577.06	18.00	-0.70	-0.15	788.55	1577.10
24.00	-0.87	-0.21	789.11	1578.23	24.00	-0.87	-0.21	789.14	1578.27
30.00	-1.01	-0.28	789.59	1579.19	30.00	-1.01	-0.28	789.63	1579.25
36.00	-1.13	-0.36	789.97	1579.95	36.00	-1.13	-0.37	790.02	1580.04
42.00	-1.22	-0.45	790.26	1580.52	42.00	-1.22	-0.46	790.31	1580.63
48.00	-1.30	-0.56	790.45	1580.90	48.00	-1.29	-0.57	790.51	1581.03
54.00	-1.35	-0.67	790.54	1581.08	54.00	-1.34	-0.68	790.62	1581.24
60.00	-1.37	-0.80	790.54	1581.08	60.00	-1.37	-0.81	790.63	1581.27

Figure 2: Kinematic characteristics of the two suspension configurations

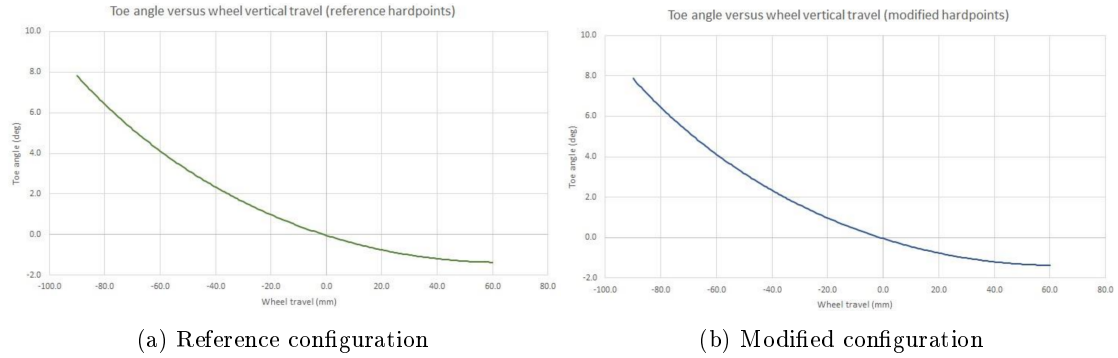


Figure 3: Toe angle vs wheel vertical travel

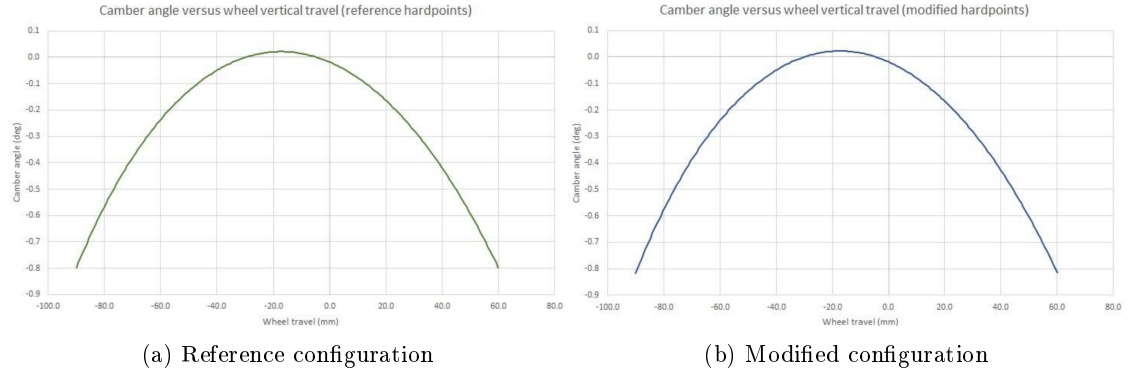


Figure 4: Camber angle vs wheel vertical travel

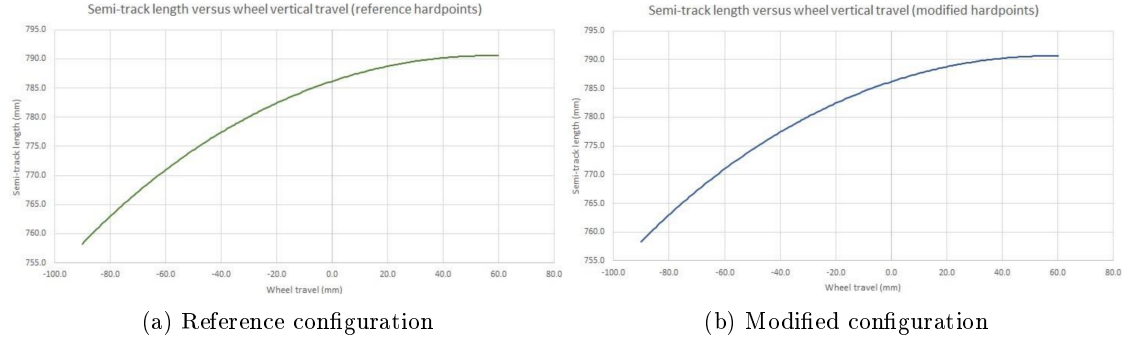


Figure 5: Semi-track length vs wheel vertical travel

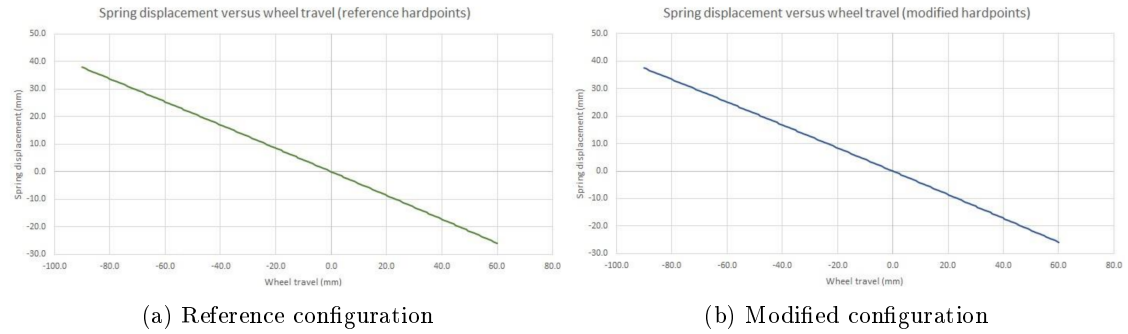


Figure 6: Spring displacement vs wheel vertical travel

5 Static forces on the suspension

Knowing the spring stiffness K_s , the spring force F_s can be calculated for each value of the displacement ξ of the spring, which is set to zero when $z = 0$, i.e. no bump or rebound is occurring, where the static load on the spring is the preload F_{s0} :

$$F_s = F_{s0} + K_s \cdot \xi \quad (1)$$

Then the ground force F_g can be calculated from the spring force F_s and displacements, using the virtual work principle:

$$F_g \cdot \delta z = F_s \cdot \delta \xi \quad (2)$$

The previous calculations apply to the linear region of the spring, but to compute the correct elastic force on the wheel caused by the suspension system, it is necessary to consider the presence of a bump stop at one suspension end, which adds non-linear stiffness to the coil spring, both in bound and rebound travel. To define the bump-stop intervention points $\xi_{b,lim}$ and $\xi_{r,lim}$, the spring maximum shortening in bump condition $\xi_{b,max}$ and its maximum lengthening in rebound

condition $\xi_{r,min}$ are retrieved from the plots of the kinematic behavior. The mean elongation of the spring ξ_m is evaluated as:

$$\xi_m = \frac{\xi_{b,max} + \xi_{r,min}}{2} \quad (3)$$

The spring's travel can be divided into three sections, assuming that the bump-stop only works in the last third. In bound conditions, these bump-stops only work between $\xi_{b,lim}$ and $\xi_{b,max}$, whereas in rebound conditions, they work between $\xi_{r,lim}$ and $\xi_{r,min}$. Where:

$$\xi_{b,lim} = \xi_{b,max} - \frac{\xi_{b,max} - \xi_m}{3} \quad (4)$$

$$\xi_{r,lim} = \xi_{r,min} - \frac{\xi_{r,min} - \xi_m}{3} \quad (5)$$

It is assumed that the bump-stops behave in the same way during bound and rebound and that the forces given by the bump-stops are:

$$F_{B,b} = K_B \cdot (\xi - \xi_{b,lim})^2 \quad (6)$$

$$F_{B,r} = K_B \cdot (\xi - \xi_{r,lim})^2 \quad (7)$$

where $\xi_{b,lim} \leq \xi \leq \xi_{b,max}$, $\xi_{r,min} \leq \xi \leq \xi_{r,lim}$ and K_b is the stiffness of the bump-stop. The stiffness K_b is determined assuming that equal force is provided by spring and bump-stop at the extremes:

$$K_S \cdot (\xi_{b,max} - \xi_m) = K_B \cdot (\xi_{b,max} - \xi_{b,lim})^2 \quad (8)$$

Therefore we can calculate the ground force which accounts for the contribution of the bump-stops, using again the virtual work principle:

$$F_G = F_{tot} \cdot \frac{\delta \xi}{\delta z} \quad (9)$$

where:

$$F_{tot} = F_S + F_B \quad (10)$$

As the last task, the ground stiffness K_g is evaluated:

$$K_G = \frac{\delta F_G}{\delta z} \quad (11)$$

The obtained results are reported in the following paragraphs.

Stiffness	K_s (N/mm)	24
Pre-load	F_{s0} (N)	4195
Mean elongation	ξ_m (mm)	5.95
Maximum bump	$\xi_{b,max}$ (mm)	37.85
Maximum rebound	$\xi_{r,min}$ (mm)	-25.96
Bump limit	$\xi_{b,lim}$ (mm)	27.22
Rebound limit	$\xi_{r,lim}$ (mm)	-15.32
Bump-stops stiffness	K_b (N/mm ²)	6.77

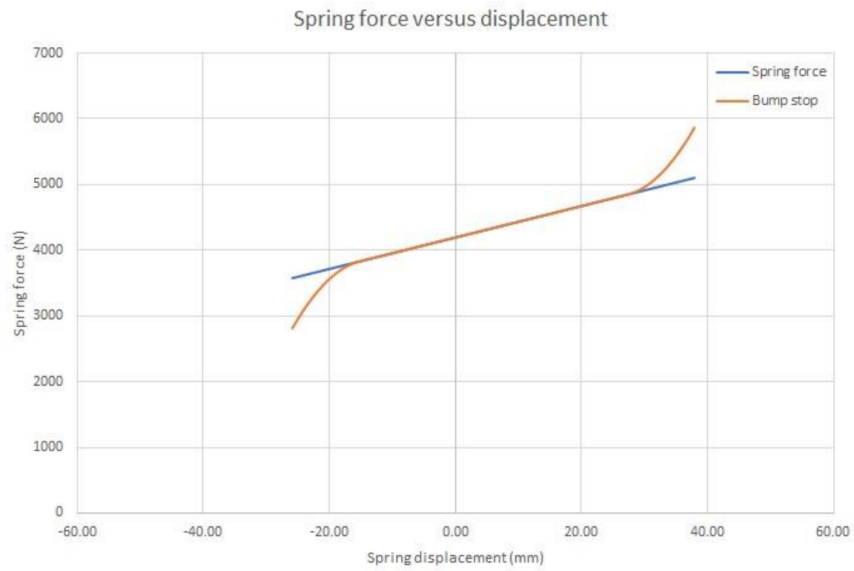


Figure 7: Spring vs force displacement

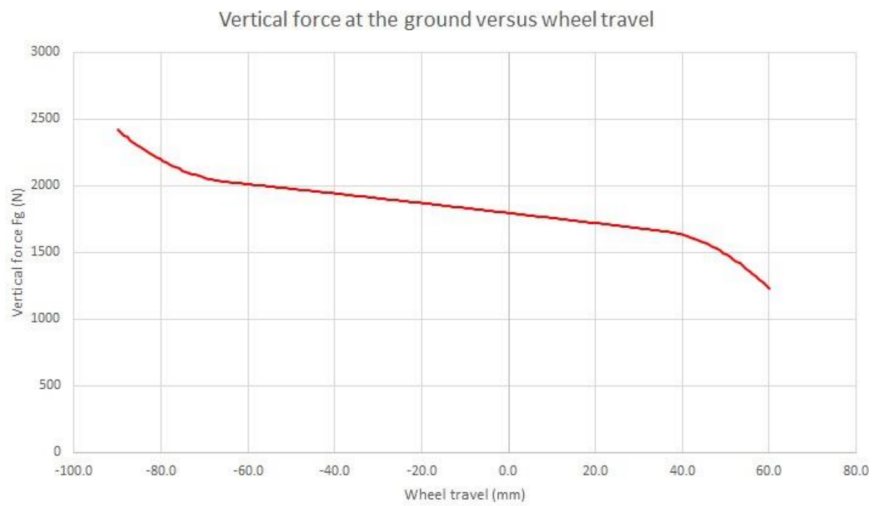


Figure 8: Vertical force at the ground vs wheel travel

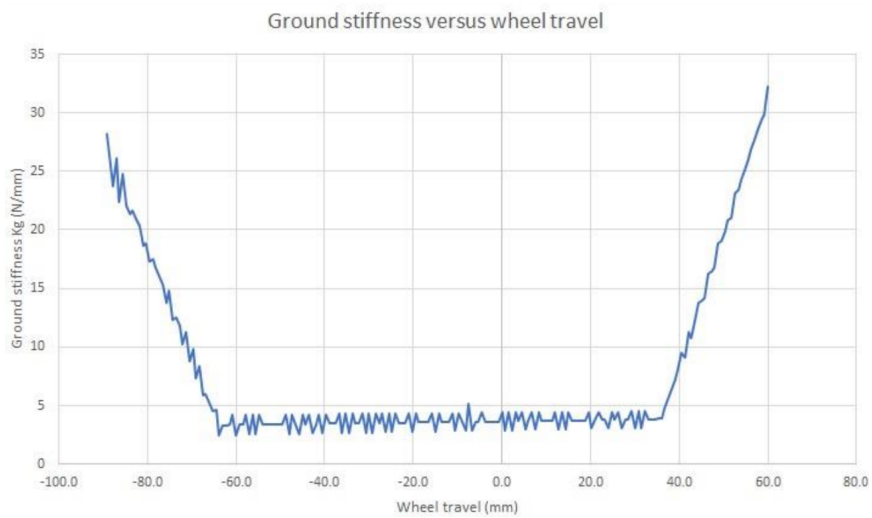


Figure 9: Ground stiffness vs wheel travel

6 Conclusions

The obtained results are reasonably accurate and coherent with the expected behavior of a double wishbone suspension. Considering a movement of the wheel going from maximum bump to maximum rebound, we get a slowly increasing toe angle and a rapid decrease of the semi-track length, while the spring displacement shows a linear behavior. The camber angle, however, has a parabolic behavior: starts from a negative value and reaches slightly above zero, to go again towards the same starting negative value. The results also show that this particular geometry has a low sensitivity to changes in the hard-points positions, since the plots of the kinematic behavior in the two cases present very small changes. As for the bump-stop design section, its implementation has produced the expected results in the force versus elongation graph as well as in the vertical force versus wheel travel graph. Moreover, the final equivalent ground stiffness values seem reasonable, with a small constant value in the middle of the spring elongation that grows as the ends are reached. The bump-stop causes a harsher behavior of the suspension around the extremities, as shown by the graphs, because it has to ensure that the maximum allowed wheel travel is not exceeded.

A Appendix

Wheel Travel	Increment	Spring displacement	Increment	Spring force	Bump-stops force	Total force	Ground force	Increment	Ground stiffness
z	δz	ξ	$\delta \xi$	Fs	FB,b,r	Ftot	Fg	δFg	Kg
(mm)	(mm)	(mm)	(mm)	(N)	(N)	(N)	(N)	(N)	(N/mm)
-90.00	-	37.85	-	5103	766	5869	2419	-	-
-89.25	-0.75	37.54	0.31	5096	722	5818	2398	21	28
-88.50	-0.75	37.24	0.31	5089	679	5768	2378	20	26
-87.75	-0.75	36.93	0.31	5081	638	5719	2360	18	24
-87.00	-0.75	36.62	0.31	5074	598	5672	2341	20	26
-86.25	-0.75	36.31	0.31	5066	559	5625	2324	17	22
-85.50	-0.75	36.00	0.31	5059	522	5581	2305	19	25
-84.75	-0.75	35.69	0.31	5051	485	5537	2289	17	22
-84.00	-0.75	35.38	0.31	5044	451	5495	2273	16	21
-83.25	-0.75	35.07	0.31	5037	417	5454	2256	16	22
-82.50	-0.75	34.76	0.31	5029	385	5414	2241	16	21
-81.75	-0.75	34.45	0.31	5022	354	5375	2225	15	20
-81.00	-0.75	34.14	0.31	5014	324	5338	2211	14	19
-80.25	-0.75	33.82	0.31	5007	295	5302	2197	14	19
-79.50	-0.75	33.51	0.31	4999	268	5268	2184	13	17
-78.75	-0.75	33.20	0.31	4992	242	5234	2171	13	17
-78.00	-0.75	32.89	0.31	4984	218	5202	2159	13	17
-77.25	-0.75	32.58	0.31	4977	195	5171	2146	12	16
-76.50	-0.75	32.27	0.31	4969	173	5142	2135	12	15
-75.75	-0.75	31.96	0.31	4962	152	5114	2125	10	14
-75.00	-0.75	31.65	0.31	4954	133	5087	2114	11	15
-74.25	-0.75	31.33	0.31	4947	115	5062	2104	9	12
-73.50	-0.75	31.02	0.31	4940	98	5037	2095	9	13
-72.75	-0.75	30.71	0.31	4932	82	5014	2086	9	12
-72.00	-0.75	30.40	0.31	4925	68	4993	2078	8	10
-71.25	-0.75	30.08	0.31	4917	56	4973	2070	8	11
-70.50	-0.75	29.77	0.31	4910	44	4954	2063	7	9
-69.75	-0.75	29.46	0.31	4902	34	4936	2056	7	10
-69.00	-0.75	29.15	0.31	4895	25	4920	2051	5	7
-68.25	-0.75	28.83	0.31	4887	18	4905	2044	6	8
-67.50	-0.75	28.52	0.31	4880	11	4891	2040	4	6
-66.75	-0.75	28.21	0.31	4872	7	4879	2035	5	6
-66.00	-0.75	27.90	0.31	4865	3	4868	2031	4	5
-65.25	-0.75	27.58	0.31	4857	1	4858	2028	3	5

-64.50	-0.75	27.27	0.31	4849	0	4849	2025	4	5
-63.75	-0.75	26.96	0.31	4842	0	4842	2023	2	2
-63.00	-0.75	26.64	0.31	4834	0	4834	2020	2	3
-62.25	-0.75	26.33	0.31	4827	0	4827	2018	3	3
-61.50	-0.75	26.02	0.31	4819	0	4819	2015	3	3
-60.75	-0.75	25.70	0.31	4812	0	4812	2012	3	4
-60.00	-0.75	25.39	0.31	4804	0	4804	2010	2	2
-59.25	-0.75	25.07	0.31	4797	0	4797	2008	3	3
-58.50	-0.75	24.76	0.31	4789	0	4789	2005	3	3
-57.75	-0.75	24.45	0.31	4782	0	4782	2002	3	4
-57.00	-0.75	24.13	0.31	4774	0	4774	2000	2	3
-56.25	-0.75	23.82	0.31	4767	0	4767	1997	3	4
-55.50	-0.75	23.50	0.31	4759	0	4759	1995	2	3
-54.75	-0.75	23.19	0.31	4752	0	4752	1992	3	4
-54.00	-0.75	22.88	0.31	4744	0	4744	1989	3	3
-53.25	-0.75	22.56	0.31	4736	0	4736	1987	3	3
-52.50	-0.75	22.25	0.31	4729	0	4729	1984	3	3
-51.75	-0.75	21.93	0.31	4721	0	4721	1982	3	3
-51.00	-0.75	21.62	0.31	4714	0	4714	1979	3	3
-50.25	-0.75	21.30	0.31	4706	0	4706	1977	3	3
-49.50	-0.75	20.99	0.32	4699	0	4699	1974	3	3
-48.75	-0.75	20.67	0.32	4691	0	4691	1971	3	4
-48.00	-0.75	20.36	0.32	4684	0	4684	1969	2	3
-47.25	-0.75	20.04	0.32	4676	0	4676	1966	3	4
-46.50	-0.75	19.73	0.32	4668	0	4668	1963	3	3
-45.75	-0.75	19.41	0.32	4661	0	4661	1961	2	3
-45.00	-0.75	19.09	0.32	4653	0	4653	1958	3	4
-44.25	-0.75	18.78	0.32	4646	0	4646	1956	3	3
-43.50	-0.75	18.46	0.32	4638	0	4638	1952	3	4
-42.75	-0.75	18.15	0.32	4631	0	4631	1950	2	3
-42.00	-0.75	17.83	0.32	4623	0	4623	1948	3	3
-41.25	-0.75	17.51	0.32	4615	0	4615	1945	3	4
-40.50	-0.75	17.20	0.32	4608	0	4608	1943	2	3
-39.75	-0.75	16.88	0.32	4600	0	4600	1939	3	4
-39.00	-0.75	16.57	0.32	4593	0	4593	1937	3	3
-38.25	-0.75	16.25	0.32	4585	0	4585	1934	3	3
-37.50	-0.75	15.93	0.32	4577	0	4577	1932	3	3
-36.75	-0.75	15.62	0.32	4570	0	4570	1928	3	4

-36.00	-0.75	15.30	0.32	4562	0	4562	1926	2	3
-35.25	-0.75	14.98	0.32	4555	0	4555	1923	3	4
-34.50	-0.75	14.67	0.32	4547	0	4547	1921	2	3
-33.75	-0.75	14.35	0.32	4539	0	4539	1918	3	4
-33.00	-0.75	14.03	0.32	4532	0	4532	1915	3	3
-32.25	-0.75	13.72	0.32	4524	0	4524	1913	3	3
-31.50	-0.75	13.40	0.32	4517	0	4517	1910	3	4
-30.75	-0.75	13.08	0.32	4509	0	4509	1908	2	3
-30.00	-0.75	12.76	0.32	4501	0	4501	1904	3	4
-29.25	-0.75	12.45	0.32	4494	0	4494	1902	2	3
-28.50	-0.75	12.13	0.32	4486	0	4486	1899	3	4
-27.75	-0.75	11.81	0.32	4478	0	4478	1896	3	4
-27.00	-0.75	11.49	0.32	4471	0	4471	1893	3	4
-26.25	-0.75	11.18	0.32	4463	0	4463	1891	2	3
-25.50	-0.75	10.86	0.32	4456	0	4456	1888	3	4
-24.75	-0.75	10.54	0.32	4448	0	4448	1886	2	3
-24.00	-0.75	10.22	0.32	4440	0	4440	1883	3	4
-23.25	-0.75	9.90	0.32	4433	0	4433	1880	3	4
-22.50	-0.75	9.59	0.32	4425	0	4425	1877	3	4
-21.75	-0.75	9.27	0.32	4417	0	4417	1875	3	4
-21.00	-0.75	8.95	0.32	4410	0	4410	1872	3	4
-20.25	-0.75	8.63	0.32	4402	0	4402	1869	2	3
-19.50	-0.75	8.31	0.32	4394	0	4394	1866	3	4
-18.75	-0.75	7.99	0.32	4387	0	4387	1864	3	4
-18.00	-0.75	7.67	0.32	4379	0	4379	1861	3	4
-17.25	-0.75	7.36	0.32	4372	0	4372	1858	3	4
-16.50	-0.75	7.04	0.32	4364	0	4364	1856	3	4
-15.75	-0.75	6.72	0.32	4356	0	4356	1852	3	4
-15.00	-0.75	6.40	0.32	4349	0	4349	1850	2	3
-14.25	-0.75	6.08	0.32	4341	0	4341	1847	3	4
-13.50	-0.75	5.76	0.32	4333	0	4333	1844	3	4
-12.75	-0.75	5.44	0.32	4326	0	4326	1842	3	4
-12.00	-0.75	5.12	0.32	4318	0	4318	1839	3	4
-11.25	-0.75	4.80	0.32	4310	0	4310	1836	3	4
-10.50	-0.75	4.48	0.32	4303	0	4303	1833	2	3
-9.75	-0.75	4.16	0.32	4295	0	4295	1830	3	4
-9.00	-0.75	3.84	0.32	4287	0	4287	1828	3	4

-8.25	-0.75	3.52	0.32	4280	0	4280	1825	2	3
-7.50	-0.75	3.20	0.32	4272	0	4272	1822	4	5
-6.75	-0.75	2.88	0.32	4264	0	4264	1819	2	3
-6.00	-0.75	2.56	0.32	4257	0	4257	1817	3	4
-5.25	-0.75	2.24	0.32	4249	0	4249	1814	3	4
-4.50	-0.75	1.92	0.32	4241	0	4241	1811	3	4
-3.75	-0.75	1.60	0.32	4233	0	4233	1808	3	4
-3.00	-0.75	1.28	0.32	4226	0	4226	1805	3	4
-2.25	-0.75	0.96	0.32	4218	0	4218	1803	3	4
-1.50	-0.75	0.64	0.32	4210	0	4210	1800	3	4
-0.75	-0.75	0.32	0.32	4203	0	4203	1797	3	4
0.00	-0.75	0.00	0.32	4195	0	4195	1794	3	4
0.75	-0.75	-0.32	0.32	4187	0	4187	1792	2	3
1.50	-0.75	-0.64	0.32	4180	0	4180	1788	3	4
2.25	-0.75	-0.96	0.32	4172	0	4172	1786	2	3
3.00	-0.75	-1.28	0.32	4164	0	4164	1783	3	4
3.75	-0.75	-1.61	0.32	4156	0	4156	1780	3	4
4.50	-0.75	-1.93	0.32	4149	0	4149	1777	3	4
5.25	-0.75	-2.25	0.32	4141	0	4141	1775	2	3
6.00	-0.75	-2.57	0.32	4133	0	4133	1772	3	4
6.75	-0.75	-2.89	0.32	4126	0	4126	1769	3	4
7.50	-0.75	-3.21	0.32	4118	0	4118	1766	2	3
8.25	-0.75	-3.53	0.32	4110	0	4110	1763	3	4
9.00	-0.75	-3.86	0.32	4102	0	4102	1760	3	4
9.75	-0.75	-4.18	0.32	4095	0	4095	1757	3	4
10.50	-0.75	-4.50	0.32	4087	0	4087	1755	3	4
11.25	-0.75	-4.82	0.32	4079	0	4079	1752	3	4
12.00	-0.75	-5.14	0.32	4072	0	4072	1749	3	4
12.75	-0.75	-5.47	0.32	4064	0	4064	1746	2	3
13.50	-0.75	-5.79	0.32	4056	0	4056	1743	3	4
14.25	-0.75	-6.11	0.32	4048	0	4048	1741	2	3
15.00	-0.75	-6.43	0.32	4041	0	4041	1737	3	4
15.75	-0.75	-6.76	0.32	4033	0	4033	1735	3	4
16.50	-0.75	-7.08	0.32	4025	0	4025	1732	3	4
17.25	-0.75	-7.40	0.32	4017	0	4017	1729	3	4
18.00	-0.75	-7.72	0.32	4010	0	4010	1726	3	4
18.75	-0.75	-8.05	0.32	4002	0	4002	1723	3	4
19.50	-0.75	-8.37	0.32	3994	0	3994	1720	3	4
20.25	-0.75	-8.69	0.32	3986	0	3986	1718	2	3
21.00	-0.75	-9.02	0.32	3979	0	3979	1715	3	4
21.75	-0.75	-9.34	0.32	3971	0	3971	1712	3	4
22.50	-0.75	-9.66	0.32	3963	0	3963	1709	3	4
23.25	-0.75	-9.99	0.32	3955	0	3955	1706	3	4
24.00	-0.75	-10.31	0.32	3948	0	3948	1704	2	3
24.75	-0.75	-10.63	0.32	3940	0	3940	1700	3	4
25.50	-0.75	-10.96	0.32	3932	0	3932	1698	3	4
26.25	-0.75	-11.28	0.32	3924	0	3924	1694	3	4
27.00	-0.75	-11.61	0.32	3916	0	3916	1692	2	3
27.75	-0.75	-11.93	0.32	3909	0	3909	1689	3	4
28.50	-0.75	-12.25	0.32	3901	0	3901	1686	3	4
29.25	-0.75	-12.58	0.32	3893	0	3893	1683	3	4
30.00	-0.75	-12.90	0.32	3885	0	3885	1681	2	3
30.75	-0.75	-13.23	0.32	3878	0	3878	1677	3	4
31.50	-0.75	-13.55	0.32	3870	0	3870	1675	2	3
32.25	-0.75	-13.88	0.32	3862	0	3862	1671	3	4
33.00	-0.75	-14.20	0.32	3854	0	3854	1669	3	4
33.75	-0.75	-14.53	0.32	3846	0	3846	1666	3	4
34.50	-0.75	-14.85	0.32	3839	0	3839	1663	3	4
35.25	-0.75	-15.18	0.33	3831	0	3831	1660	3	4
36.00	-0.75	-15.50	0.33	3823	0	3823	1657	3	4
36.75	-0.75	-15.83	0.33	3815	-2	3813	1654	4	5

37.50	-0.75	-16.15	0.33	3807	-5	3803	1649	4	6
38.25	-0.75	-16.48	0.33	3800	-9	3790	1645	5	6
39.00	-0.75	-16.80	0.33	3792	-15	3777	1639	5	7
39.75	-0.75	-17.13	0.33	3784	-22	3762	1633	6	8
40.50	-0.75	-17.45	0.33	3776	-31	3745	1626	7	10
41.25	-0.75	-17.78	0.33	3768	-41	3727	1619	7	9
42.00	-0.75	-18.11	0.33	3760	-53	3708	1611	8	11
42.75	-0.75	-18.43	0.33	3753	-66	3687	1603	8	11
43.50	-0.75	-18.76	0.33	3745	-80	3665	1593	9	12
44.25	-0.75	-19.08	0.33	3737	-96	3641	1583	10	14
45.00	-0.75	-19.41	0.33	3729	-113	3616	1573	10	14
45.75	-0.75	-19.74	0.33	3721	-132	3589	1562	11	14
46.50	-0.75	-20.06	0.33	3713	-152	3561	1550	12	16
47.25	-0.75	-20.39	0.33	3706	-174	3532	1537	12	17
48.00	-0.75	-20.72	0.33	3698	-197	3501	1525	13	17
48.75	-0.75	-21.04	0.33	3690	-222	3468	1511	14	19
49.50	-0.75	-21.37	0.33	3682	-248	3434	1496	14	19
50.25	-0.75	-21.70	0.33	3674	-275	3399	1482	15	20
51.00	-0.75	-22.02	0.33	3666	-304	3362	1466	16	21
51.75	-0.75	-22.35	0.33	3659	-335	3324	1450	16	21
52.50	-0.75	-22.68	0.33	3651	-366	3284	1433	17	23
53.25	-0.75	-23.01	0.33	3643	-400	3243	1415	18	23
54.00	-0.75	-23.33	0.33	3635	-435	3200	1397	18	24
54.75	-0.75	-23.66	0.33	3627	-471	3156	1378	19	25
55.50	-0.75	-23.99	0.33	3619	-509	3111	1359	19	26
56.25	-0.75	-24.32	0.33	3611	-548	3064	1339	20	27
57.00	-0.75	-24.64	0.33	3604	-588	3015	1318	21	28
57.75	-0.75	-24.97	0.33	3596	-631	2965	1296	21	29
58.50	-0.75	-25.30	0.33	3588	-674	2914	1274	22	29
59.25	-0.75	-25.63	0.33	3580	-719	2861	1252	22	30
60.00	-0.75	-25.96	0.33	3572	-766	2806	1228	24	32